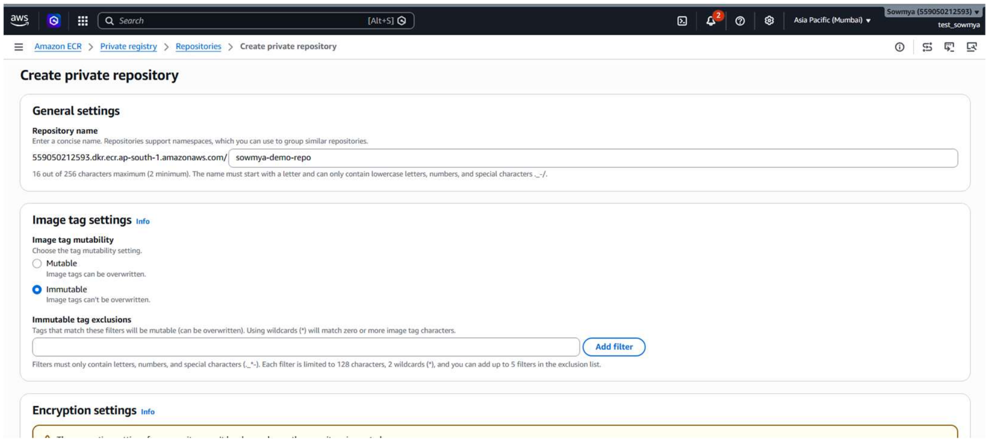
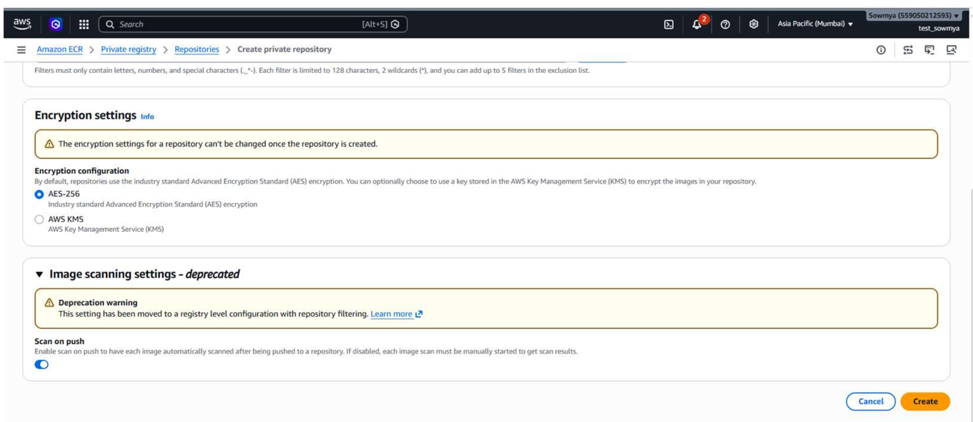
Demonstration of ECR and ECS

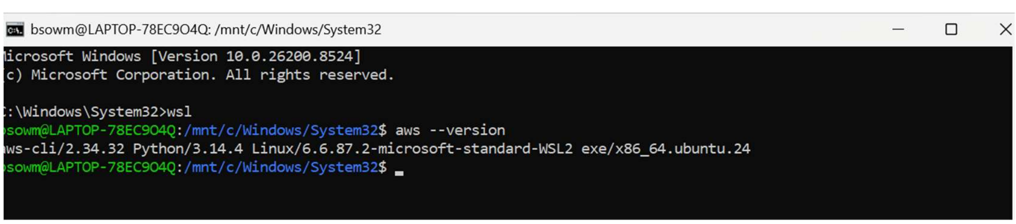
Step-1: Login to your AWS IAM user account

Step-2: Go to ECR->click on create repository





Step-3: Open command prompt and check if cli is installed or not using command aws –version



Step-4: Now connect to aws using command aws configure. Then after install docker inside wsl following below commands

sudo apt update

sudo apt install docker.io -y

docker –version

Step-5: start running the docker using below command

sudo service docker start

sudo docker ps

Step-6: Instead of using sudo everytime lets fix the permission using below command

sudo usermod -aG docker $USER

Then after try running docker ps. If its works correctly permissions are fixed.

Step-6: create a directory and then create a html file as below

mkdir ~/ecr-demo

cd ~/ecr-demo

nano index.html

cat index.html

```
bsowm@LAPTOP-78EC9O4Q: ~/ecr-demo
  GNU nano 7.2                                         index.html *
<!DOCTYPE html>
<html>
<body>
<h1>Hello from ECR!</h1>
</body>
</html>
```

Step-7: Now create a docker file, build image, test image locally.

nano Dockerfile

FROM nginx:latest

COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80

Save and exit Dockerfile after entering the above contents.

docker build -t sowmya-demo-repo .

docker images

```
bsowm@LAPTOP-78EC904Q:~/ecr-demo$ docker images
                                                                         i Info →   U  In Use
IMAGE                      ID             DISK USAGE   CONTENT SIZE   EXTRA
nginx:latest               5aca99593157       241MB            66MB
sowmya-demo-repo:latest    dad44226ad7d       238MB          63.1MB
```

docker run -d -p 8080:80 sowmya-demo-repo

docker ps

open browser and paste http://localhost:8080

Step-8: Login to ECR and push the image to ECR using below commands.

aws ecr get-login-password --region ap-south-1 | docker login --username AWS --password-stdin 559050212593.dkr.ecr.ap-south-1.amazonaws.com

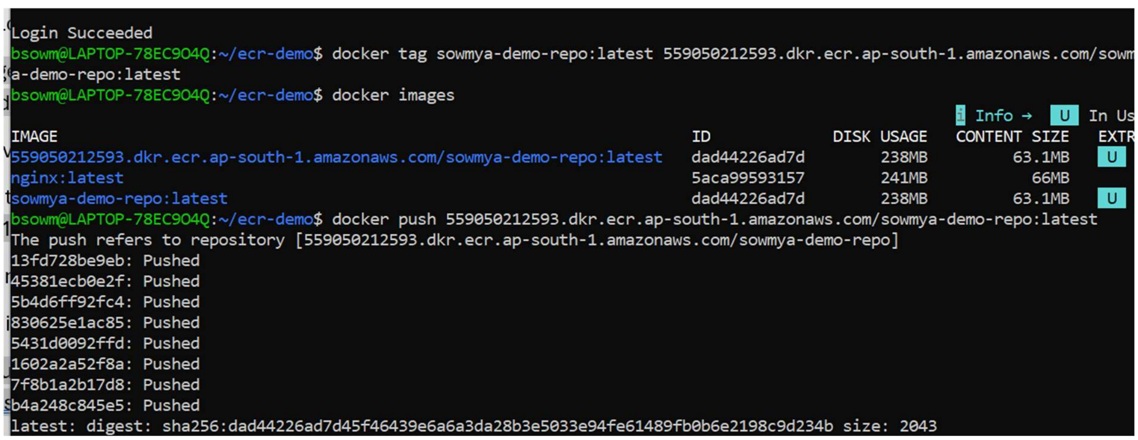
Here I have given my credentials to login.

Then Tag the image using docker tag sowmya-demo-repo:latest 559050212593.dkr.ecr.ap-south-1.amazonaws.com/sowmya-demo-repo:latest

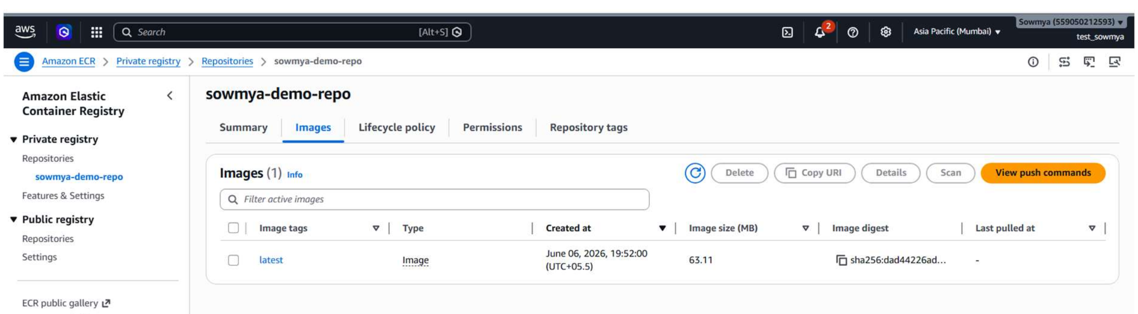
Verify using docker images you will see the images created.

Push the image to ECR using command

docker push 559050212593.dkr.ecr.ap-south-1.amazonaws.com/sowmya-demo-repo:latest



Step-9: Go to AWS console->ECR->open the repository created you will able to see the image pushed.



ECS

Architecture flow diagram

Developer

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Docker Image

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Amazon ECR

↓

ECS Task Definition

↓

ECS Service (Fargate)

↓

Public IP

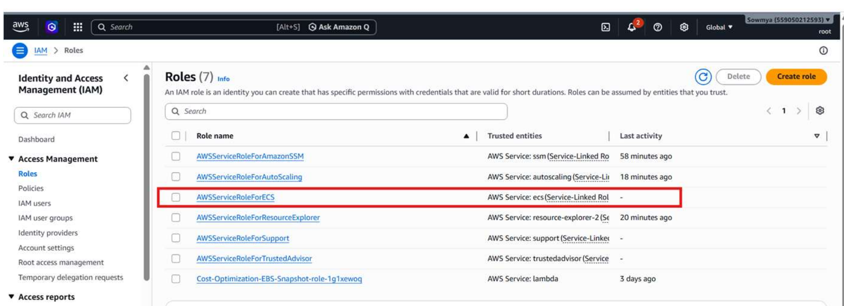
↓

Web Application

Step-1: Go to AWS console->ECS before that verify if you have valid IAM roles i.e AWSServiceRoleForECS. You can also add this via wsl with following commands

```
aws iam create-service-linked-role --aws-service-name ecs.amazonaws.com
```

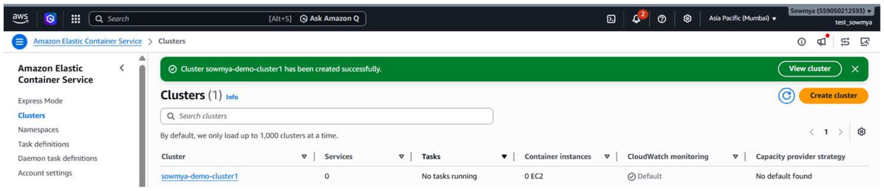
Run only if AWSServiceRoleForECS does not already exist.

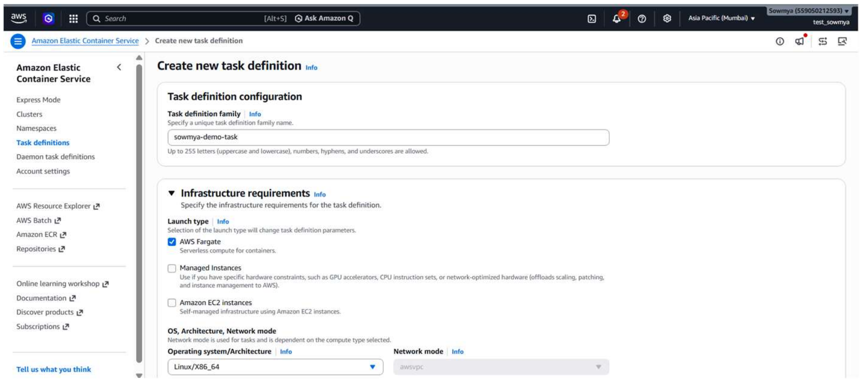


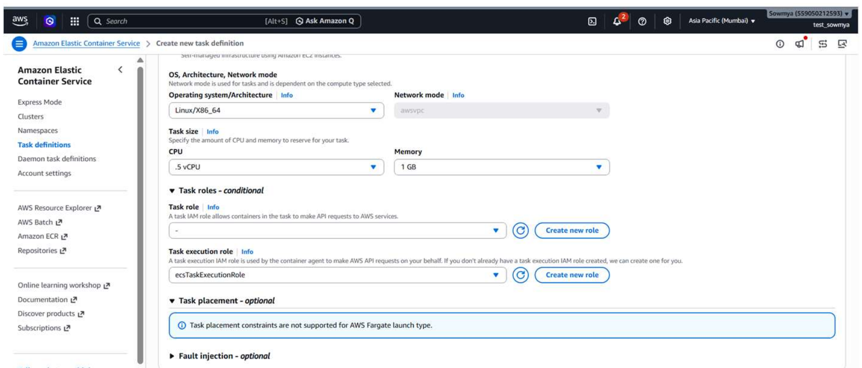
Step-2: Create a cluster and task definition along with Task Execution Role: ecsTaskExecutionRole

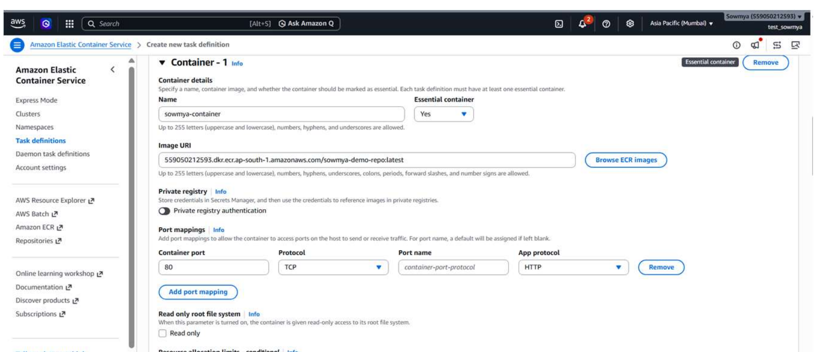
This role allows ECS tasks to pull images from ECR and send logs to CloudWatch.

Also, Assign Public IP = Enabled

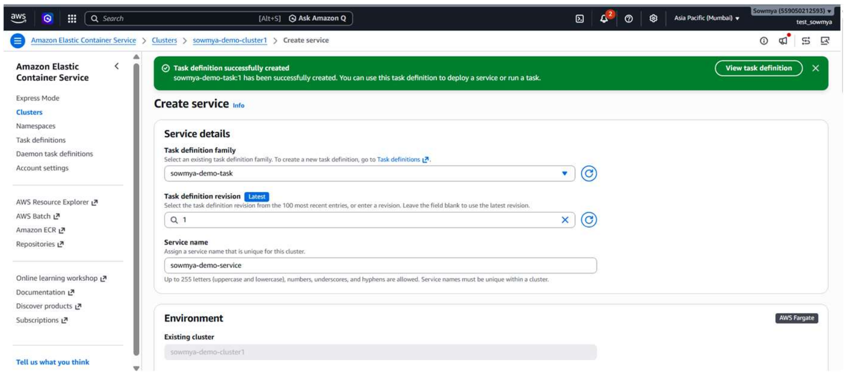


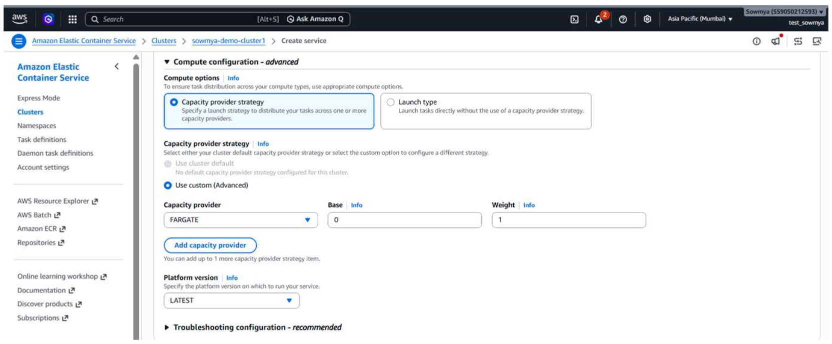


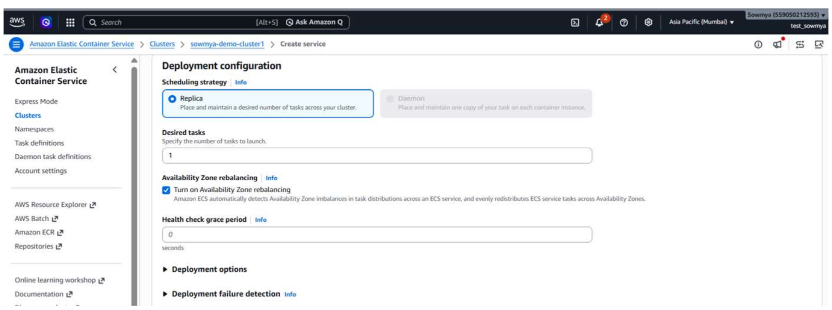


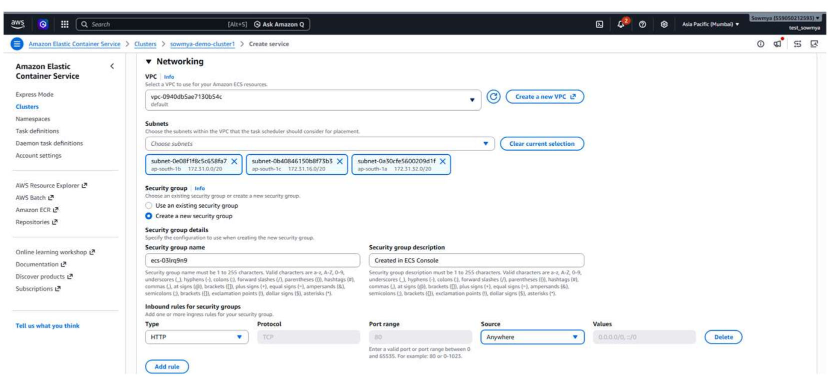


Step-3: Go to the cluster you have created and you will services. Click on create.

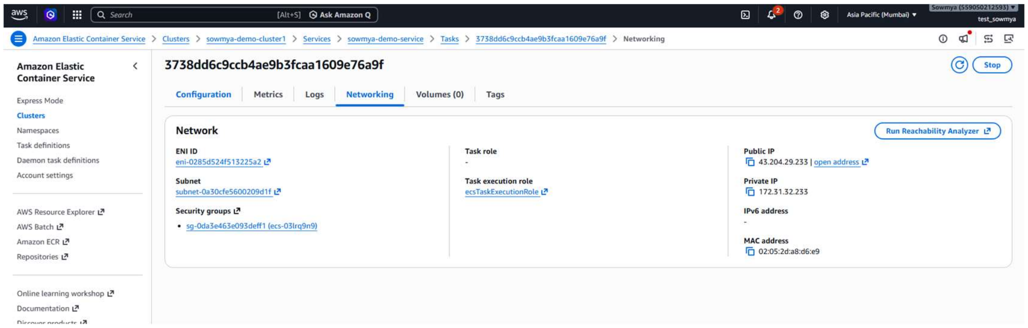


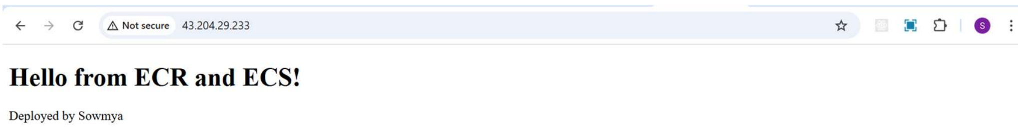






Step-4: Once service is created successfully go to service->select service created->Tasks->networking and copy the public IP and paste in web browser you will be able to access the page successfully.





Now, do not forget to delete the services once completed.

Cleanup

1. Delete ECS Service
2. Delete ECS Cluster
3. Delete ECR Repository
4. Deregister Task Definitions
5. Delete CloudWatch Log Groups (optional)